

Placement Empowerment Program
Cloud Computing and DevOps Centre

**Host a Static Website Locally: Set Up a Local Server Apache and
Host a Simple HTML page with your name**

Name: Neithal Pillai

Department : IT

Introduction and Overview

This Proof of Concept (POC) aims to provide a comprehensive understanding of how to locally host a static website using the Apache HTTP Server. The process involves installing and configuring an Apache server, ensuring that it is properly set up to serve web pages, and hosting a basic HTML file. By following these steps, you will develop hands-on experience in working with Apache, which is one of the most widely used web servers globally. Understanding how to set up a local server is a fundamental skill for web developers and system administrators, as it forms the foundation for web hosting, server configuration, and website deployment.

Objective

The primary objective of this project is to:

1. **Install and Set Up an Apache Web Server Locally** – Learn how to install and configure the Apache HTTP Server on your system, ensuring it runs correctly.
2. **Configure the Server to Serve Static Web Pages** – Understand the necessary configurations required for hosting static files and making them accessible through a web browser.
3. **Create and Host a Simple HTML Page** – Develop a basic HTML file that contains your name and successfully host it on the locally configured Apache server.

Importance of Local Hosting

Local hosting plays a vital role in web development, offering a secure and controlled environment for testing and experimentation. It is an essential skill for developers, as it allows them to work on websites and applications without the need for immediate internet connectivity or live deployment. Some of the key advantages of local hosting include:

- **Hands-On Learning** – Setting up a local server helps in gaining practical experience with web server configurations, file hosting, and troubleshooting. This foundational knowledge is valuable for anyone pursuing web development or system administration.
- **Testing and Debugging in a Safe Environment** – By hosting a website locally, developers can test and debug their code without affecting a live

site. This reduces the risk of errors impacting real users and allows thorough testing before deployment.

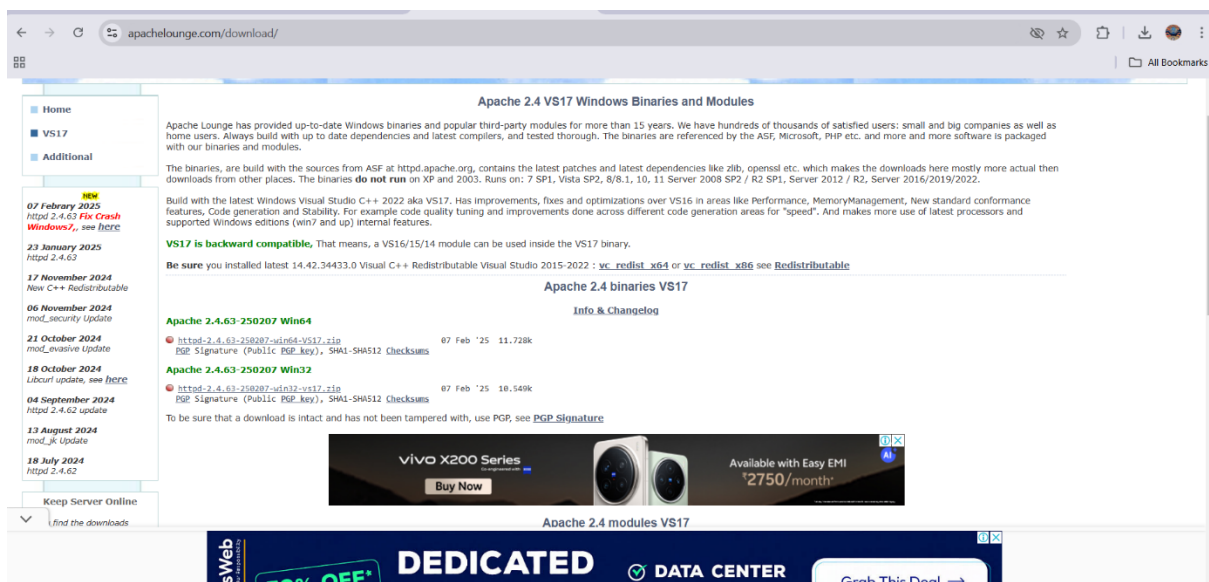
- **Offline Web Development** – A local web server enables developers to work on their projects without an active internet connection. This is particularly useful for those working in areas with limited connectivity or for individuals who prefer to build and test their applications offline before making them publicly available.

By completing this POC, you will gain the necessary skills to set up a fully functional local web server using Apache, making it easier to test, develop, and host static web content efficiently.

Steps for execution:

◆ Step 1: Downloading and Extracting Apache

I started by downloading the Apache HTTP Server (Win64 version) from the official website. After downloading, I extracted the files into a folder on my desktop. Later, I moved the Apache folder to a different location inside my Downloads folder, which caused some configuration issues later.



◆ Step 2: Installing Apache as a Service

To install Apache as a service, I opened Command Prompt as an administrator and tried running the installation command. However, I immediately ran into an issue where the system did not recognize the command.

I realized that I had not navigated to the correct directory where the Apache executable file was located. Once I changed the directory to the correct path and retried the command, the installation was successful.

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>httpd.exe -k restart
```

```
httpd.exe: Syntax error on line 39 of C:/Users/neith/Downloads/httpd-2.4.63-250207-win64-VS17 (1)/Apache24/conf/httpd.conf: ServerRoot must be a valid directory
```

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>httpd.exe -k restart
```

```
AH00526: Syntax error on line 252 of C:/Users/neith/Downloads/httpd-2.4.63-250207-win64-VS17 (1)/Apache24/conf/httpd.conf:
```

```
DocumentRoot 'C:/Apache24/htdocs' is not a directory, or is not readable
```

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>httpd.exe -k restart
```

```
AH00558: httpd.exe: Could not reliably determine the server's fully qualified domain name, using fe80::bc76:2f2:b7b8:9075. Set the 'ServerName' directive globally to suppress this message
```

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>httpd.exe -k restart
```

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>http://localhost/
```

```
'http:' is not recognized as an internal or external command,  
operable program or batch file.
```

◆ Step 3: Fixing Configuration Errors

After successfully installing Apache, I tried starting the service, but I encountered a configuration error related to the ServerRoot directory.

I opened the configuration file and checked the line where the error was occurring. The issue was that the path still pointed to the old directory where Apache was originally extracted. Since I had moved the folder, I had to update this path in the configuration file.

After fixing this and restarting Apache, I encountered another error stating that the DocumentRoot directory was missing or unreadable. This again was due to the path not being updated after moving Apache to a new location. I corrected the path in the configuration file and restarted Apache.

Next, I got a warning about Apache not being able to determine the server's fully qualified domain name. To fix this, I updated the ServerName directive in the configuration file and restarted the server.

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>httpd.exe -k restart  
httpd.exe: Syntax error on line 39 of C:/Users/neith/Downloads/httpd-2.4.63-250207-win64-VS17  
(1)/Apache24/conf/httpd.conf: ServerRoot must be a valid directory
```

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>httpd.exe -k restart  
AH00526: Syntax error on line 252 of C:/Users/neith/Downloads/httpd-2.4.63-250207-win64-VS17  
(1)/Apache24/conf/httpd.conf:  
DocumentRoot 'C:/Apache24/htdocs' is not a directory, or is not readable
```

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>httpd.exe -k restart  
AH00558: httpd.exe: Could not reliably determine the server's fully qualified domain name, using  
fe80::bc76:2f2:b7b8:9075. Set the 'ServerName' directive globally to suppress this message
```

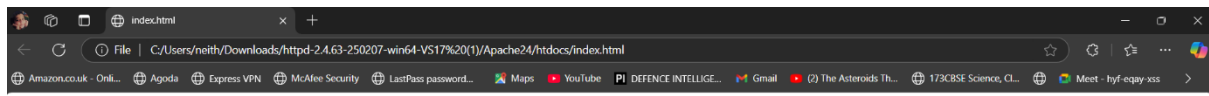
```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>httpd.exe -k restart
```

```
C:\Users\neith\Downloads\httpd-2.4.63-250207-win64-VS17 (1)\Apache24\bin>http://localhost/  
'http:' is not recognized as an internal or external command,  
operable program or batch file.
```

◆ Step 4: Hosting a Simple Static Website

Once Apache was running properly, I navigated to the folder where web files were supposed to be placed. I created a simple HTML file with my name and saved it in this location.

I restarted Apache to apply the changes and then opened my web browser. When I typed "localhost" in the address bar, my HTML page loaded successfully, confirming that Apache was serving the website correctly.



Neithal Pillai Dept: IT- A (SJCE)

I have successfully made a locally hosted website using Apache24



Key Takeaways

- I had to ensure that the correct paths were set in the configuration file after moving Apache.
- Running commands from the right directory was important to avoid errors.
- Restarting Apache was necessary after making changes to the configuration.
- Using "localhost" in the browser allowed me to test my website.
- Running Command Prompt as an administrator helped in executing service-related commands.